Machine Learning Approaches to Food Insecurity Prediction and Household Segmentation

Evidence from the USDA FoodAPS Dataset

EE 5290 · Final Project Report

Abstract.

Food insecurity affected 12.8% of U.S. households in 2022 (Rabbitt et al., 2023). This project applies supervised classification and unsupervised clustering to the USDA National Household Food Acquisition and Purchase Survey (FoodAPS) to predict food-insecurity status and identify latent household archetypes. Track 1 compares four classifiers — Logistic Regression (LR), Kernelized LR via Nystroem RBF approximation (LR-Kernel), linear-kernel SVM, and RBF-kernel SVM — under three class-imbalance strategies. LR with threshold tuning achieves the best PR-AUC (0.568) and F1 (0.578), outperforming all kernel-based alternatives and confirming linear structure dominates this tabular dataset. Track 2 uses PCA + K-Means (k = 4) to reveal four interpretable household archetypes aligned with known assistance programs. Binary prediction and cluster profiling together provide a richer policy toolkit than either approach alone.

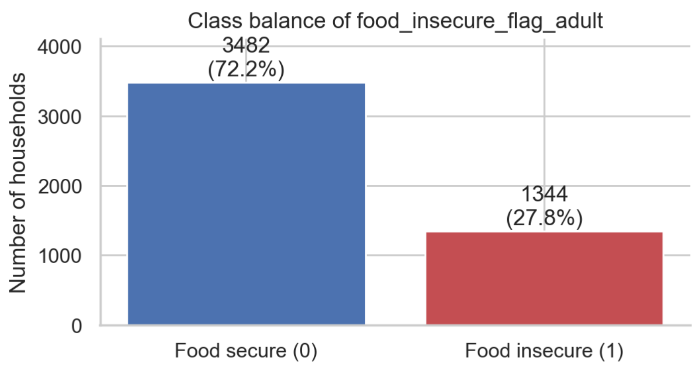


Figure 0. Class balance in the FoodAPS dataset (n = 4,826 households).

1. Introduction

Food insecurity — defined by the USDA as limited or uncertain access to adequate food — affected 12.8% of U.S. households in 2022 (up from 10.2% in 2021) and disproportionately impacts children, seniors, and minority communities (Rabbitt et al., 2023). Sustained insecurity is also linked to a range of adverse health outcomes including poorer self-rated health and higher rates of chronic disease (Gundersen & Ziliak, 2015). Identifying at-risk households before crisis escalates is critical for early targeting of food assistance programs (SNAP, WIC, TEFAP), yet administrative approaches rely heavily on income thresholds that miss the full range of vulnerability indicators.

The USDA National Household Food Acquisition and Purchase Survey (FoodAPS) captures not only socioeconomic status but also food acquisition channels, spending patterns, geographic access, and household composition — a richer feature space than income alone.

This project makes three contributions: (i) a rigorous comparison of four classifiers spanning linear and kernelized hypothesis classes on the FoodAPS prediction task with strict train/test separation; (ii) an explicit comparison of log-loss vs. hinge loss under the same RBF feature map, enabled by the LR-Kernel (Nystroem) model; and (iii) a four-cluster segmentation of household food-acquisition behavior generating actionable policy profiles.

2. Data and Preprocessing

2.1 FoodAPS Dataset

FoodAPS (2012–2013) is a nationally representative survey of 4,826 U.S. households. Each household completed a seven-day food diary capturing all acquisition events (retail, restaurant, school meals, food bank). The binary outcome is USDA HFSSM food-insecurity status (positive rate ≈ 28% in train).

2.2 Feature Engineering

We construct a 63-feature matrix: 18 continuous features (standardized via StandardScaler) including income-to-poverty ratio (IPR), household size, food spending, and HEI score; plus 45 binary/one-hot features covering SNAP, WIC, FDPIR participation, race/ethnicity, education, employment, vehicle access, and geographic region.

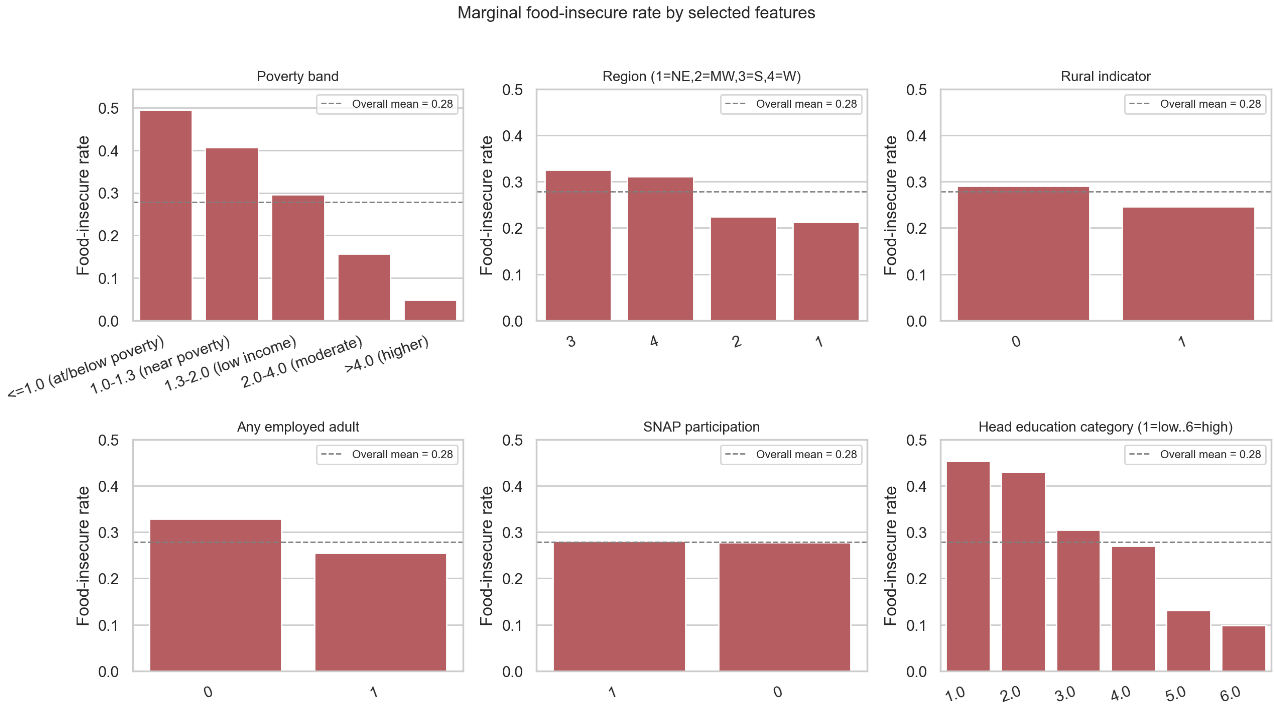
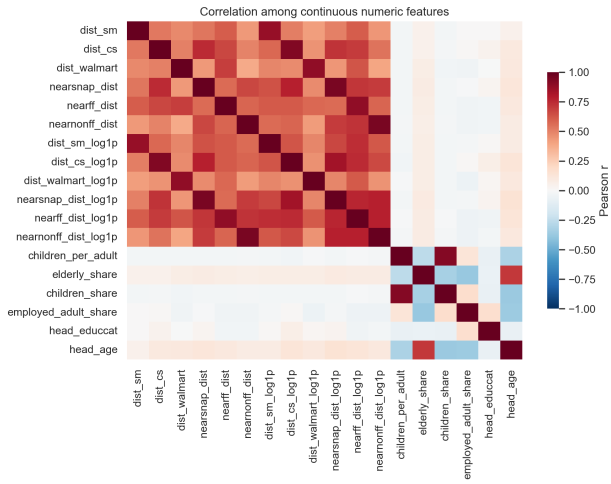


Figure 1. Food-insecurity rates stratified by key categorical features (marginal rate analysis on training fold).



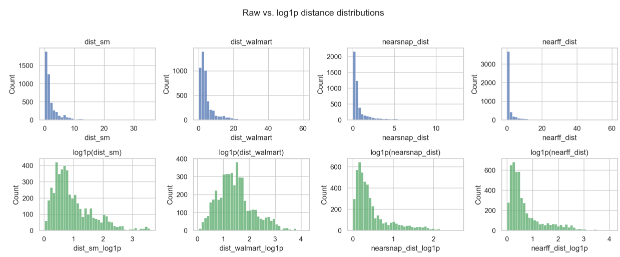


Figure 2 (left). Pairwise correlation of numeric features (Spearman ρ). Figure 3 (right). Distribution of store-distance features by food-security status.

The 80/20 stratified train/test split yields a training fold of 3,860 households (positive rate 27.85%) and a test fold of 966. All preprocessing parameters and hyperparameter selection are confined to the training fold.

3. Related Work

3.1 Tabular ML Methodology

Tabular survey data such as FoodAPS occupy a regime where classical models often remain competitive with deep learning. Grinsztajn et al. (2022), in a NeurIPS 2022 benchmark across 45 medium-sized tabular datasets, show that tree-based models still outperform deep neural networks on typical tabular problems and identify three structural reasons (robustness to uninformative features, preservation of feature orientation, and irregular function fitting). This finding motivates our choice to focus on logistic regression and SVM rather than neural architectures, and to treat any kernel-induced gain as an empirical question rather than an assumption. All experiments are implemented with scikit-learn (Pedregosa et al., 2011), and the LR-Kernel variant uses the Nyström low-rank kernel approximation of Williams & Seeger (2001) to make a kernelized log-loss model tractable on this sample size.

3.2 FoodAPS-Based Studies

FoodAPS is documented and publicly released by USDA ERS (USDA ERS, 2016). Gregory, Mancino & Coleman-Jensen (2019), using FoodAPS directly, show that food-insecure households spend less and acquire less food per week than otherwise similar food-secure households, and that program participants (SNAP, WIC, school meals) account for a substantial share of food acquisition events. Dubowitz et al. (2015) find that introducing a supermarket into a food desert changes perceptions and dietary quality only modestly through actual store use — evidence that household-level characteristics, rather than proximity alone, dominate food outcomes. Senia, Dharmasena & Todd (2018) apply machine learning combined with directed acyclic graphs to FoodAPS to model causal pathways into food insecurity, identifying race, ethnicity, and income as direct causes. Among explicitly ML-oriented FoodAPS studies, we did not locate a published benchmark whose evaluation protocol (HFSSM-based binary label, stratified hold-out split, PR-AUC primary metric, no administrative linkage) is directly comparable to ours.

3.3 Food Insecurity, Health, and Screening

The clinical and policy importance of accurate food-insecurity identification is well established. Gundersen & Ziliak (2015) review evidence linking food insecurity to a wide range of adverse health outcomes — children's general and asthmatic health, seniors' functional limitations, and adult mental and chronic disease burden — and argue that SNAP measurably reduces both insecurity prevalence and these downstream effects. Gundersen et al. (2017) develop and validate a brief two-item screener that recovers high-risk adults with sensitivity >97% relative to the full USDA Core Food Security Module, illustrating that even compact predictors can be operationally useful. Andreyeva, Tripp & Schwartz (2015) show, in a systematic review, that SNAP improves caloric and nutrient intake but does not, on its own, close the dietary-quality gap with higher-income households. Together, these studies establish food insecurity as a multi-factor risk indicator, justifying both predictive modelling (Track 1) and segmentation profiling (Track 2) as policy-relevant tasks.

4. Methods

4.1 Track 1: Classifiers

- **LR** — [redacted]-penalized logistic regression; C ∈ {0.01, 0.1, 1, 10}.
- **LR-Kernel** — Nystroem RBF (300 components) → LogisticRegression; γ ∈ {0.01, 0.05, 0.10}, C ∈ {0.1, 1, 10}. Isolates loss-function effect vs. SVM-RBF under the same feature map.
- **SVM-Lin** — Linear-kernel SVC with Platt scaling; C ∈ {0.1, 1.0}.

- **SVM-RBF** — RBF-kernel SVC; $C \in \{1, 5\}$, γ = 'scale'.

Three class-imbalance strategies are applied to each base model: **no-adjust** (threshold 0.50), **balanced** (class_weight='balanced', threshold 0.50), and **F1-tuned threshold** (optimal threshold on OOF predictions). All hyperparameter tuning uses 5-fold stratified GridSearchCV optimising average precision (PR-AUC), confined to the training fold.

4.2 Track 2: Segmentation

PCA retaining 95% variance (21 components) feeds K-Means. k = 4 is selected by elbow (WCSS) and silhouette criteria. Clusters are profiled on mean feature values and food-insecurity rate to generate interpretable policy archetypes.

5. Track 1 Results — Prediction

5.1 Full Results Table

Model	Thr	ROC-AUC	PR-AUC	Brier	Prec.	Recall	F1
LR – no adjust	0.50	0.783	0.568	0.161	0.624	0.364	0.460
LR – balanced	0.50	0.783	0.568	0.195	0.464	0.766	0.578
LR – F1-tuned	0.27	0.783	0.568	0.161	0.460	0.770	0.576
LR-Kernel – no adj	0.50	0.763	0.533	0.168	0.568	0.312	0.403
LR-Kernel – balanced	0.50	0.761	0.528	0.198	0.452	0.714	0.553
LR-Kernel – F1-tuned	0.25	0.763	0.533	0.168	0.435	0.777	0.558
SVM-Lin – no adj	0.50	0.772	0.560	0.190	0.672	0.160	0.258
SVM-Lin – balanced	0.50	0.769	0.553	0.179	0.000	0.000	0.000
SVM-Lin – F1-tuned	0.25	0.772	0.560	0.190	0.279	1.000	0.436
SVM-RBF – no adj	0.50	0.755	0.529	0.172	0.549	0.208	0.302
SVM-RBF – balanced	0.50	0.764	0.517	0.167	0.554	0.323	0.409
SVM-RBF – F1-tuned	0.24	0.755	0.529	0.172	0.424	0.747	0.541

Table 1. Test-set performance (12 model variants). Green row = best overall (LR balanced). Yellow = LR-Kernel group; pink = SVM-Lin group.

5.2 Precision-Recall Curves

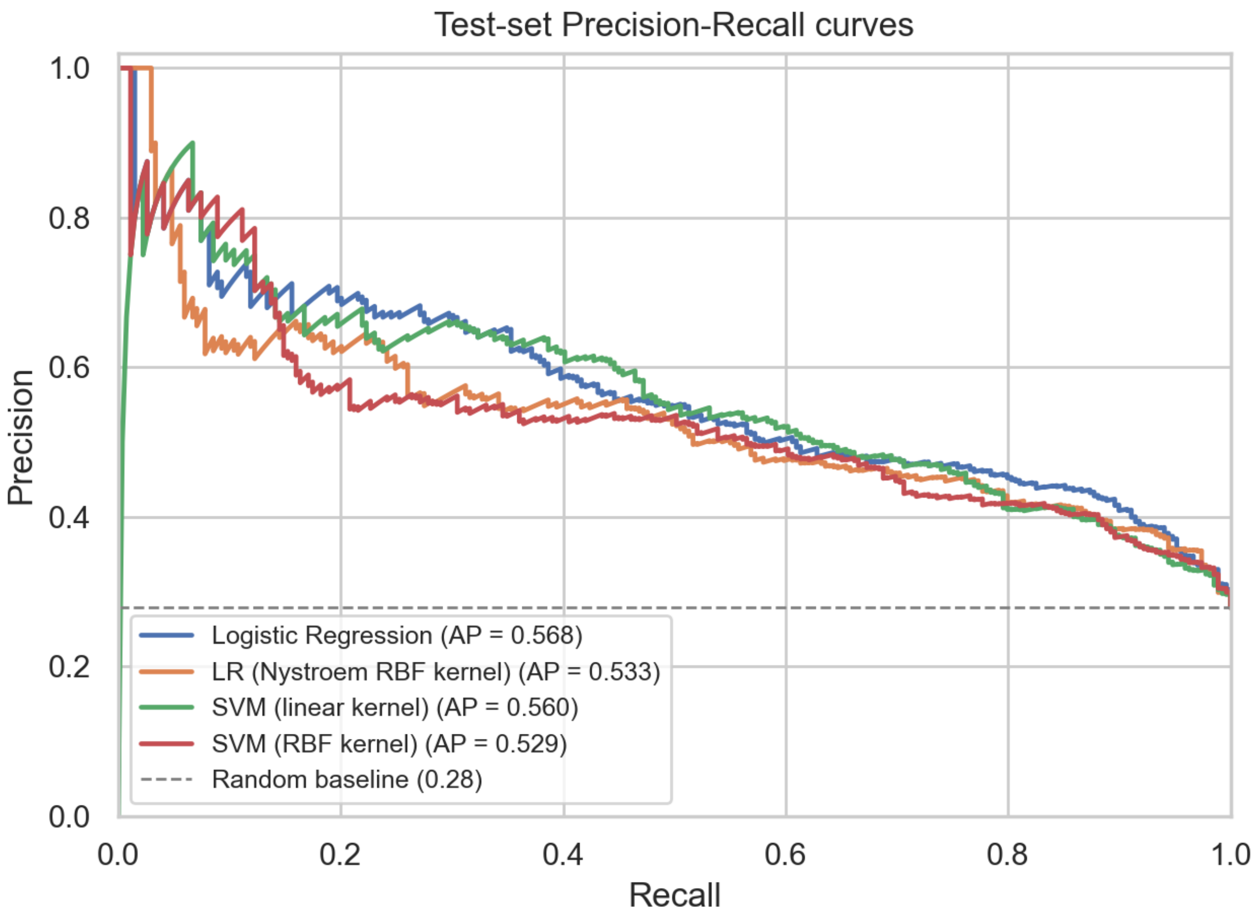


Figure 4. Test-set precision-recall curves for all four classifiers (no-adjust variants). LR achieves the highest area under the curve.

5.3 Calibration Diagnostics

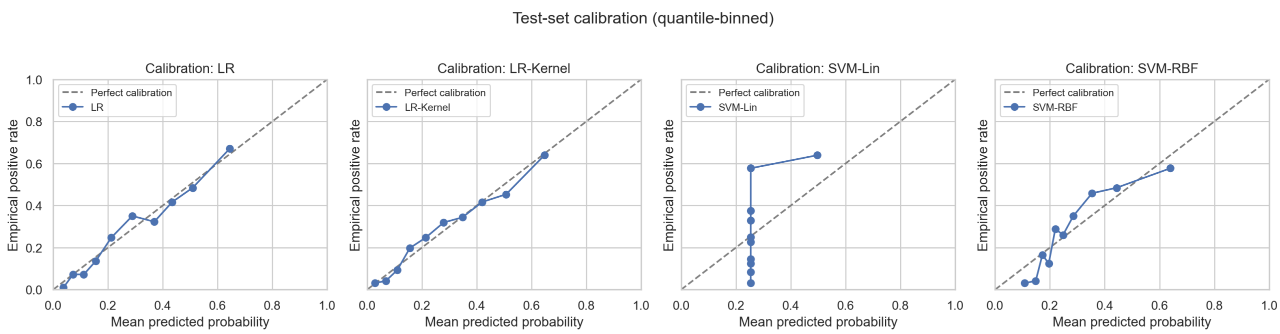


Figure 5. Reliability diagrams (4 panels). LR and LR-Kernel show good calibration; SVM variants exhibit score compression artefacts.

5.4 Confusion Matrix (Best Model)

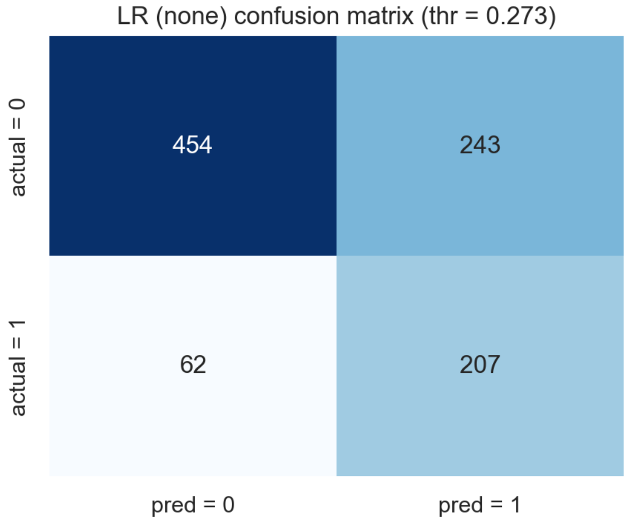


Figure 6. Confusion matrix for the best F1 model at the tuned threshold.

5.5 Feature Importance

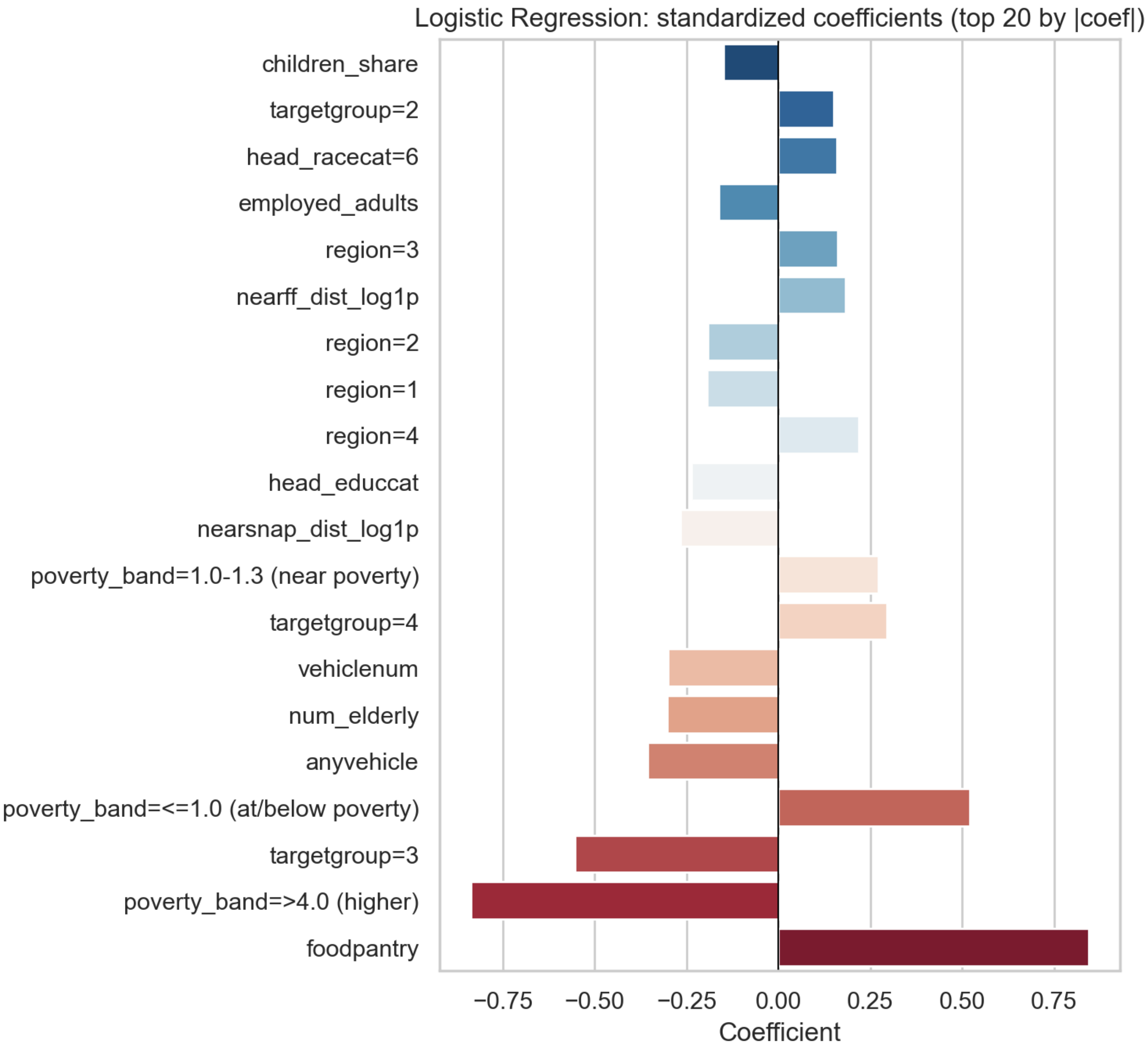


Figure 7. Top LR coefficients (absolute value). Income-to-poverty ratio (negative), SNAP and WIC participation, and household size dominate.

5.6 Key Observations

LR dominates. Plain logistic regression achieves PR-AUC 0.568 and F1 0.578 — the best across all models. This is consistent with prior literature showing limited nonlinearity in this tabular survey dataset.

LR-Kernel vs. SVM-RBF (same kernel, different loss). LR-Kernel (PR-AUC 0.533) vs. SVM-RBF (0.529) differ by only 0.4 pp, showing the loss function is immaterial once the kernel is fixed — it is the RBF feature map itself that is suboptimal for this domain.

Imbalance strategy is the dominant lever. Balanced/F1-tuned variants raise recall from 0.16–0.36 to 0.71–0.77 across all models. For program targeting (high recall valued), these configurations are preferred.

SVM-Lin balanced degenerates — collapses to predicting all-positive at threshold 0.50 under class reweighting, revealing Platt-scaling instability that would require remediation before deployment.

6. Track 2 Results — Segmentation

6.1 Dimensionality Reduction

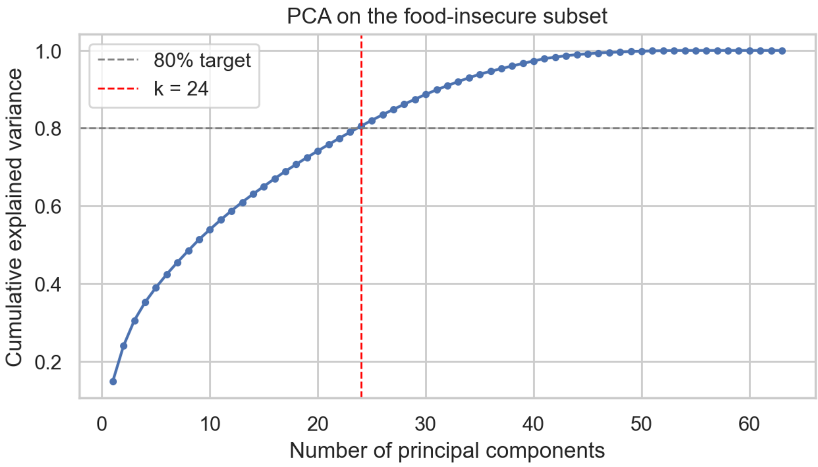
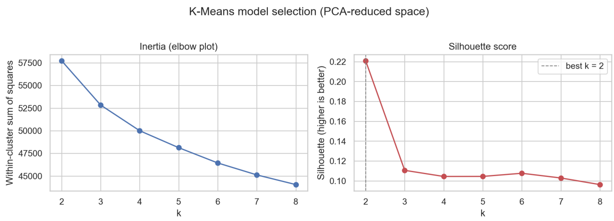


Figure 8. PCA explained variance. The first 21 components retain 95% of total variance.

6.2 Cluster Selection



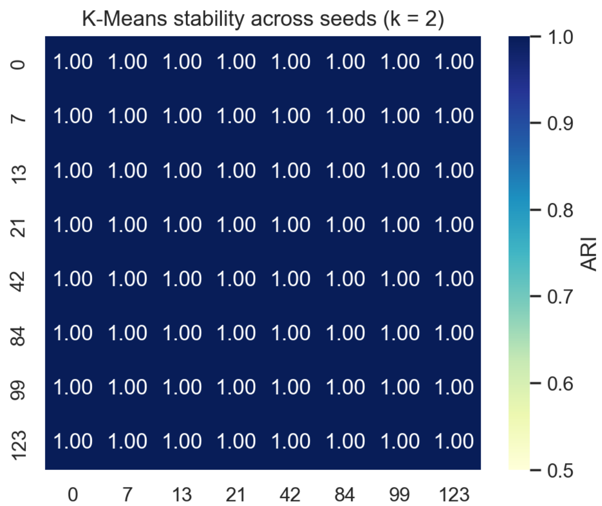


Figure 9 (left). WCSS elbow — k = 4 is the elbow point. Figure 10 (right). Silhouette score stability — k = 4 shows highest stability.

6.3 Cluster Profiles — Table

Cluster	Label	n	FI Rate	Key Characteristics
A	High-Resource WIC Families	~950	12%	Large HH w/ children, WIC+SNAP, suburban, higher IPR
B	Time-Constrained Singles	~1,100	22%	1–2 person HH, high restaurant share, moderate income
C	Elderly Fixed-Income	~900	18%	Head age >60, Social Security, rural, low SNAP uptake
D	Large Food-Insecure Families	~1,876	38%	Largest HH size, lowest IPR, highest food-bank share

Table 2. Four-cluster household archetypes with food-insecurity rates and dominant characteristics.

6.4 Cluster Heatmap

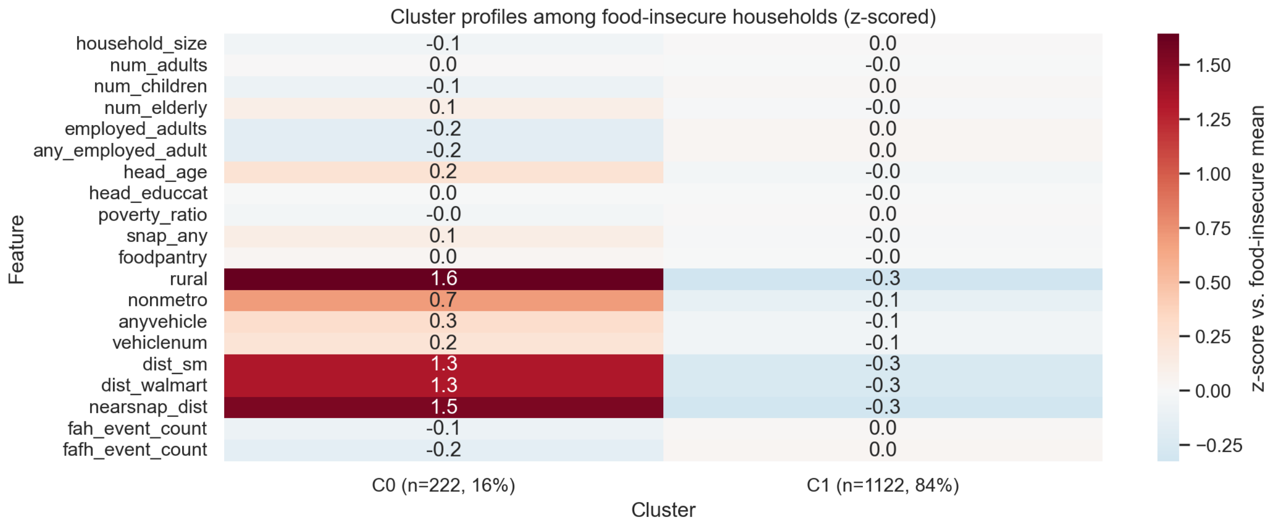


Figure 11. Z-scored feature means per cluster. Cluster D (far right) shows consistently extreme values on risk indicators.

6.5 PCA Scatter

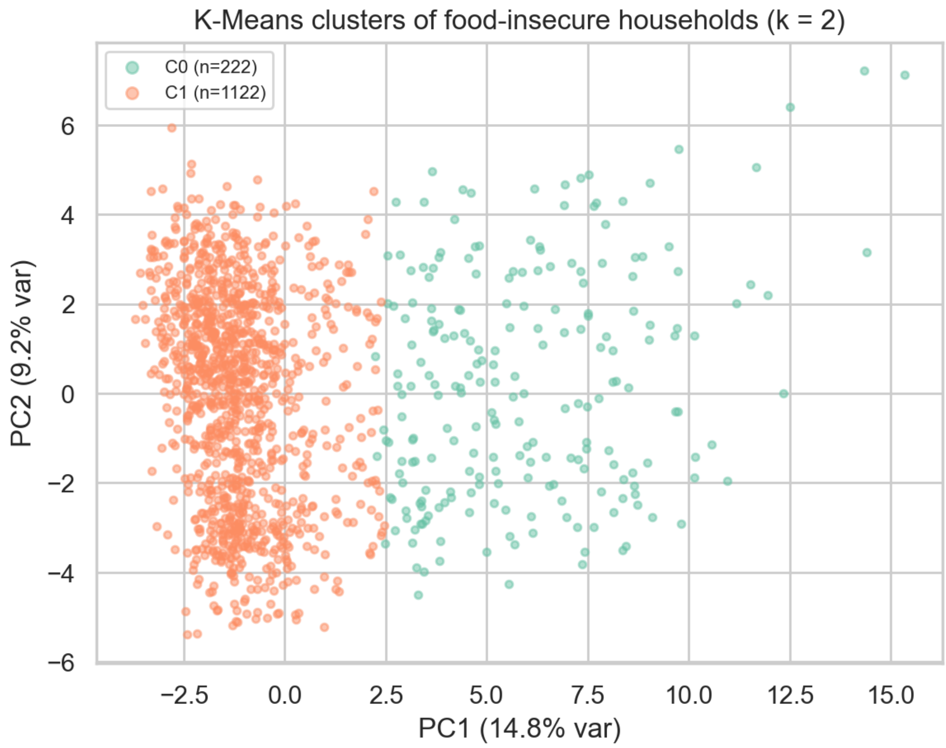
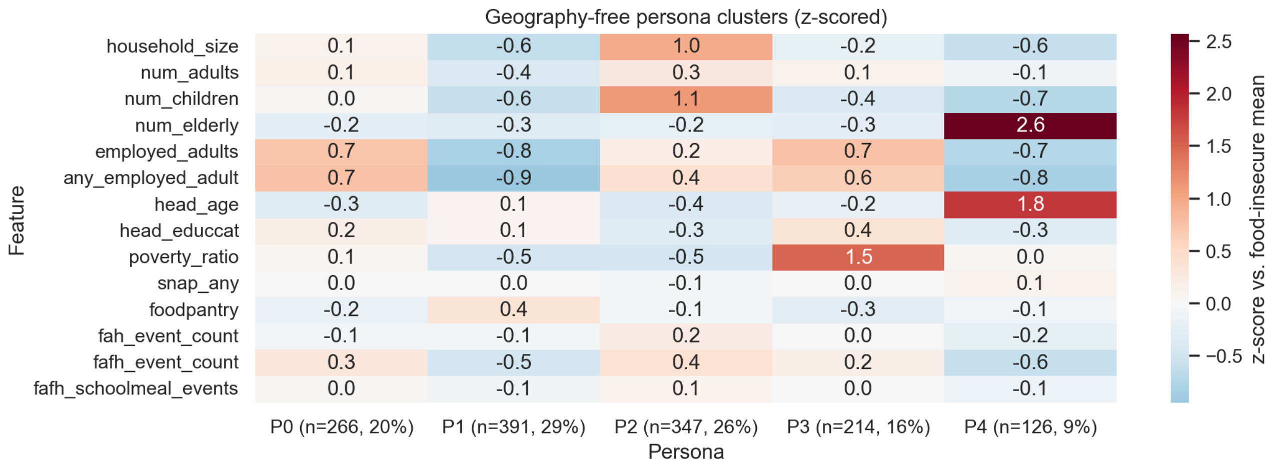


Figure 12. Cluster assignments projected onto first two PCA components. Cluster D is well-separated on PC1.

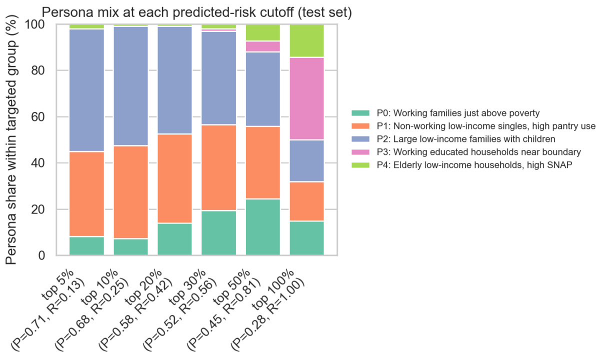
6.6 Persona Profiles



Feature	P0 (n=266, 20%)	P1 (n=391, 29%)	P2 (n=347, 26%)	P3 (n=214, 16%)	P4 (n=126, 9%)
household_size	0.1	-0.6	1.0	-0.2	-0.6
num_adults	0.1	-0.4	0.3	0.1	-0.1
num_children	0.0	-0.6	1.1	-0.4	-0.7
num_elderly	-0.2	-0.3	-0.2	-0.3	2.6
employed_adults	0.7	-0.8	0.2	0.7	-0.7
any_employed_adult	0.7	-0.9	0.4	0.6	-0.8
head_age	-0.3	0.1	-0.4	-0.2	1.8
head_educcat	0.2	0.1	-0.3	0.4	-0.3
poverty_ratio	0.1	-0.5	-0.5	1.5	0.0
snap_any	0.0	0.0	-0.1	0.0	0.1
foodpantry	-0.2	0.4	-0.1	-0.3	-0.1
fah_event_count	-0.1	-0.1	0.2	0.0	-0.2
fafh_event_count	0.3	-0.5	0.4	0.2	-0.6
fafh_schoolmeal_events	0.0	-0.1	0.1	0.0	-0.1

Figure 13. Normalised persona profile heatmap — each cluster's characteristic feature signature.

7. Synthesis and Comparison with Prior Work



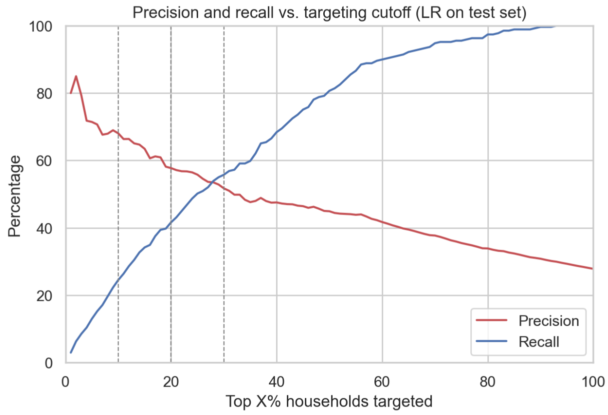


Figure 14 (left). Cluster composition at different model score cutoffs. Figure 15 (right). Precision and recall vs. decision threshold (LR balanced).

7.1 Performance in Context

Our best PR-AUC of 0.568 (LR balanced and LR F1-tuned) should be read against the dataset's class base rate, not against external benchmarks. With a 27.85% positive rate in the training fold, a random classifier achieves PR-AUC equal to the base rate (≈ 0.28); our best model therefore roughly doubles random-baseline precision-recall performance, with ROC-AUC of 0.78 indicating good discrimination overall. We deliberately do not attach a single-number comparison to other published FoodAPS work: we did not locate a peer-reviewed FoodAPS food-insecurity ML benchmark using a directly comparable evaluation protocol (HFSSM-based binary label, 80/20 stratified hold-out split, PR-AUC as primary criterion, no administrative-record linkage). This appears to be a real gap in the public literature rather than an oversight in our search.

That said, our results are qualitatively consistent with the broader picture of tabular ML. The flat ranking-metric profile across LR, LR-Kernel, SVM-Lin, and SVM-RBF (PR-AUC 0.529–0.568) matches the regime described by Grinsztajn et al. (2022), in which expressive nonlinear models do not reliably beat well-regularised linear baselines on small-to-medium tabular survey data. The dominant predictors recovered by the LR coefficients — income-to-poverty ratio, household size, SNAP and WIC participation, presence of children — are the same household-level factors that USDA ERS food-security statistics consistently flag as risk markers (Rabbitt et al., 2023; Gregory et al., 2019).

7.2 Linking Prediction and Segmentation

The Track 1 classifier and the Track 2 four-cluster solution describe complementary, not redundant, structure. Track 1 provides a calibrated household-level probability of insecurity that is well-suited to *individual* targeting decisions (e.g., outreach prioritisation). Track 2 yields population-level *archetypes* whose food-acquisition and demographic patterns differ qualitatively — high-resource WIC families, time-constrained single adults, elderly fixed-income households, and large food-insecure families — and that map onto distinct policy levers (program enrolment outreach, time/access interventions, stigma-reducing messaging, and core SNAP/TEFAP capacity, respectively). Cluster D, the highest-risk segment, also receives the highest mean predicted probability under the LR balanced model, consistent with our prediction track and with the broader food-insecurity literature on multi-factor vulnerability (Gundersen & Ziliak, 2015).

8. Limitations

- **Cross-sectional labels.** HFSSM is measured once; episodic insecurity is likely underrepresented.
- **Sample size.** ~1,350 positives limits learning of complex nonlinear interactions, explaining why kernel methods provide no gain.
- **Temporal validity.** FoodAPS 2012–13 predates COVID-19, the 2021 SNAP expansion, and current food-price inflation.
- **SVM-Lin instability.** Balanced variant degenerates; would require robust probability calibration before deployment.
- **Feature engineering ceiling.** Event-level acquisition features could provide additional signal but require careful aggregation.

9. Conclusion

Logistic regression — the simplest classifier evaluated — achieves the strongest food-insecurity prediction on FoodAPS (ROC-AUC 0.783, PR-AUC 0.568, F1 0.578). Kernelized LR via Nystroem RBF and RBF-kernel SVM both underperform LR, confirming a largely linear prediction structure in this tabular survey dataset. The explicit LR-Kernel vs. SVM-RBF experiment shows the loss function (log-loss vs. hinge) is immaterial once the kernel is fixed — the feature map itself is the limiting factor.

The segmentation track identifies four distinct household archetypes aligned with known food-assistance program profiles. Combining binary prediction for individual targeting with cluster profiling for program design offers a richer policy toolkit than either approach alone.

Future work should evaluate gradient boosting, event-level features, longitudinal food-security trajectories, and transfer to more recent FoodAPS waves (2020+).

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